Microstructure Features

- Definition:
  - "the process and outcomes of exchanging assets under explicit trading rules", O'Hara(1995)
- microstructure data:
  - order cancellation, double auction book, queues,
    partial fills (incomplete order completion), correction, replacement
    - queues: effect order but not yet cancelled

- main source would be *FIX*(financial information eXchange) messages
  - The level of detail contained in FIX messages provides researchers with the ability to understand how market participants conceal and reveal their intentions.
  - microstructural data one of the most important ingredients for building predictive ML features.

Main types of orders

1. **market orders**:
    - specifies a quantity to be sold or bought at whatever price
2. **Limit orders**:
    - specifies a quantity to be sold(bought) at a limit price above(below)

3. **Stop orders**:

- Like a limit order but with inverted limits,
- speicifying a quantity to be sold (bought) and a limit price below(above).
- If the price goes below(above) a certain limit,
  the asset is sold(bought) to 'stop' losses

Main trading systems: trading

1. Order driven:

- Traders place orders before prices are set either by market makers or by a centralized mechanism or auction
- Trading can be continuous or in batches at discrete intervals
- Batch auction to open continuous trading

2. Quote driven

- Market makers set bid and ask prices and traders submit orders
- Continuous dealer market: a trader can get immediate execution from the market maker

Market maker's adverse selection problem

- Market makers face an adverse selection problem as traders may possess private information on return
- Order driven system has a signalling flavor since the informed traders move first
- Quote driven system has a screening flavor since the uninformed party moves first

See "Information and Learning in Markets: The Impact of Market Microstructure" by Xavier Vives (2010), pp 110

Literature Review

- 1st generation: price data
- 2nd generation: price, volume
- 3rd generation: price, volume, private information feature(imposing a strategic or sequential trading process)

1st generation

1. tick rule

2. Roll's model

3. High-low volatility

4. Corwin and Schultz

1. tick rule

- buyer(seller) initiated price change sets to 1(-1)

$$b_t = \begin{cases} 1, & \Delta p_t > 0 \\ -1, & \Delta p_t < 0 \\ b_{t-1}, & \Delta p_t = 0 \end{cases}$$

where $b_0 = 1, t = 1, 2, \cdots, T$

- Even if tick rule is simple, it achieved huge empirical success with transformation accompanied.
  - using Kalman filter on $E_t[b_{t+j}]$ for $j \geq 1$
  - considering structural breaks on prediction
  - treating b_t as entropy
  - using Wald-Wolfowitz's tests
  - using fractional differential model

2. Roll's model

- How does BAS(bid-ask spread) affect on trades?
- Let $m_t = \frac{p_t^b + p_t^a}{2}$ be mid-price and assume the following:

$$m_t = m_{t-1} + u_t$$
$$p_t = m_t + b_t c,$$

where $u_t : iidN(0, \sigma_u^2), c = (BAS/2), b_t \in \{-1, 1\}$

- 2nd moments could be expressed as BAS

$$\Delta p_t = u_t + c(b_t - b_{t-1})$$
$$\Delta p_t^2 = u_t^2 + c^2(b_t^2 + b_{t-1}^2 - 2b_t b_{t-1})$$
$$E(\Delta p_t^2) = \sigma_u^2 + 2c^2 \quad \left(\because var(b_t) = 1, E(b_t b_{t-1}) = 0, \text{with } P(b_t = 1) = 1/2\right)$$
$$E(\Delta p_t \Delta p_{t-1}) = -c^2$$

- the otherway around interpretation
  - the bid-ask spread is a function of the serial covariance of price changes
  - the true (unobserved) price's noise(u_t) is a function of the observed price change($E\left(\Delta p_t^2\right)$)
    and the serial covariance of price changes($E\left(\Delta p_t \Delta p_{t-1}\right)$).

- Using Roll's estimates, we can derive informative features regarding the **market's liquidity** conditions.

$$c = (\text{effective BAS}) = \sqrt{max\Big(E(\Delta p_t \Delta p_{t-1}), 0\Big)}$$

$$\sigma_u^2 = E(\Delta p_t^2) + 2c^2$$

2-1: Extended Roll's model

- Model:

$$
\begin{aligned}
p_t &= m_t + cb_t \\
m_t &= m_{t-1} + \omega_t \\
\omega_t &= \lambda b_t + v_t
\end{aligned}
$$

$$
p_t = m_{t-1} + (\lambda + c) b_t + v_t
$$

- What makes the above model different from the original Roll model?
  - BAS: composed of two cost components (may be interpreted as inventory and aymmetric information cost)

$$\begin{cases} p_t^B = m_{t-1} - (\lambda + c) + v_t, & \text{if } \ b_t = -1 \\ p_t^A = m_{t-1} + (\lambda + c) + v_t, & \text{if } \ b_t = 1 \end{cases}$$

$$\Longrightarrow (p_t^A - p_t^B) = 2(\lambda + c)$$

- How to interpret $(\lambda + c)$?
  - commonly interpret v_t =**(public news shock)**, λ=**(private information related item)**
  - (public news arrived; u_t)

$$\Longrightarrow \text{(Dealer or Market Maker set BAS)}$$

$$\Longrightarrow \text{(trading happend; } q_t\text{)}$$

$$\Longrightarrow \text{(update efficient price; } m_t\text{)}$$

- Empirical aspects of E-Roll model

$$\Delta p_t = -cb_{t-1} + (\lambda + c)b_t + v_t$$
$$\implies E(\Delta p_t^2) = c^2 + (c+\lambda)^2 + \sigma_v^2 \text{ :volatility reflects private information}$$
$$\implies E(\Delta p_t \Delta p_{t-1}) = -c(c+\lambda)$$

- How to set up empirical testing with above? (you may skip):
  - now we have two equations($\gamma_0 = E(\Delta p_t^2), \gamma_1 = E(\Delta p_t \Delta p_{t-1})$, however, three parameters(λ, c, σ_v^2): identification problem
  - adding one more equation: $Var(w_t) = \lambda^2 + \sigma_v^2 = \gamma_0 + 2\gamma_1$

- Still we need more assumption on $Var(w_t)$
  $\Longrightarrow Var(w_t) = \frac{Var(m_t - m_{t-k})}{k} \approx \frac{Var(p_t - p_{t-k})}{k}$
  - additional assumption says that the variance of the real price is multipled by the length of the interval
    and microstructure effects dissipates over the longer period

> For more on Bas or private information see **'Hasbrouck' or 'Bas'**

3. High-Low Volatility

- Parkinson(1980) the following volatility formulae when underlying asset follows *geometric brownian motion*
  - geometric brownian motion: $dp_t = \mu dt + \sigma p_t dw_t$
  - $E\left[\frac{1}{T}\sum_{t=1}^{T}\left(log\left[\frac{H_t}{L_t}\right]\right)^2\right] = k_1\sigma_{HL}, \ \text{where} \ \ k_1 = 4log[2]$
  - $E\left[\frac{1}{T}\sum_{t=1}^{T}\left(log\left[\frac{H_t}{L_t}\right]\right)^2\right] = k_2\sigma_{HL} \ \ \text{where} \ \ k_2 = \sqrt{\frac{8}{\pi}}$
- σ_{HL} can be estimated using high-low equity prices if underlying asset shows very handy distribution

4. Corwin-Schultz BAS proxy

- Corwin and Schultz(2012) introduced BAS estimator from high and low prices
  - high(low) prices are *almost always* matched against the offer(bid). Hence the ratio of H/L reflects fundamental volatility and BAS
  - H/L induced by volatility increases proportionately with the time elapsed b/w two observations

- CS estimator

$$100 \times \frac{BAS_t}{p_t} \equiv S_t = \frac{2(e^{\alpha_t} - 1)}{1 + e^{\alpha_t}}$$

$$\text{where } \alpha_t = \frac{\sqrt{2\beta_t} - \sqrt{\beta_t}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma_t}{3 - 2\sqrt{2}}},$$

$$\beta_t = E\left[\sum_{j=0}^{1} \left(log\left[\frac{H_{t-j}}{L_{t-j}}\right]\right)^2\right],$$

$$\gamma_t = \left(log\left[\frac{H_{t-1,t}}{L_{t-1,t}}\right]\right)^2,$$

$H_{t-1,t}$=high price over 2 bars

- CS's BAS estimator is "helpful" where no order book existed, such as corporate bond market

Second Generation: Strategic Trade Models

- Second generation models focusing on illiquidity that is an very important informative feature in ML models
- Strategic interaction b/w informed and uninformed traders. Hence, these models focusing on signed volume and order flow imbalance
- **microstructural estimates' t-values are sometimes more informative than the estimates themselves.**

Kyle (1985)

- market maker; informed traders; noise traders
- market features:
  - a risky asset with terminal value $v = N(p, \Sigma)$
  - noise trader: trades quantity $u = N(0, \sigma_u^2)$
  - informed trader knowing v and demands a quantity x through market order
  - total order flow $y = x + u$
  - market maker(mm)
    - observe order y and set price p
    - can't distinguish b/w where orders coming from
    - mm adjust prices responding to the order flow imbalance.
      Hence, positive relation b/w price change and order flow imbalance, called **market impact**

- informed trader conjecture that mm has a linear price adjustment fucntion
  $p = \lambda y + \mu \ \ \text{where } \lambda \text{ is an inverse measure of liquidity.}$
- mm conjectures that informed trader's demand is a linear fucntion of v
  $x = \alpha + \beta v$

- informed traders' profit maximization

$$\pi = (v - p)x \quad \text{s.t. } p = \lambda y + \mu,\ y = x + u$$
$$\Rightarrow x = \frac{v - \mu - \lambda u}{2\lambda} \quad (\text{in random setting, u disappears as E(u)=0})$$

- Hence the optimal trading rule for informed trader is a linear trading rule (if mm use the linear price setting rule):

$$x = \alpha + \beta v, \quad \text{where } \alpha = -\frac{\mu}{2\lambda}, \beta = \frac{1}{2\lambda}$$

- Informed traders can make profit thru the following three sources
  - The security's mispricing.
  - The variance of the noise trader's net order flow: The higher the noise, the easier the informed trader can conceal his intentions.
  - The reciprocal of the terminal security's variance. The lower the volatility, the easier to monetize the mispricing.

- In sum, interpretation of Kyle's λ
  - it can be interpreted as an inverse measure of liquidity.
    - That is, the lower is λ, the more liquid is the asset, and the greater is the volume required to move the market makers' prices.
  - 1/λ measures the depth of the market
  - A parameter λ determined endogenously within the model

- how to estimate with real data?

$$\Delta p_t = \lambda(b_t V_t) + \epsilon_t$$

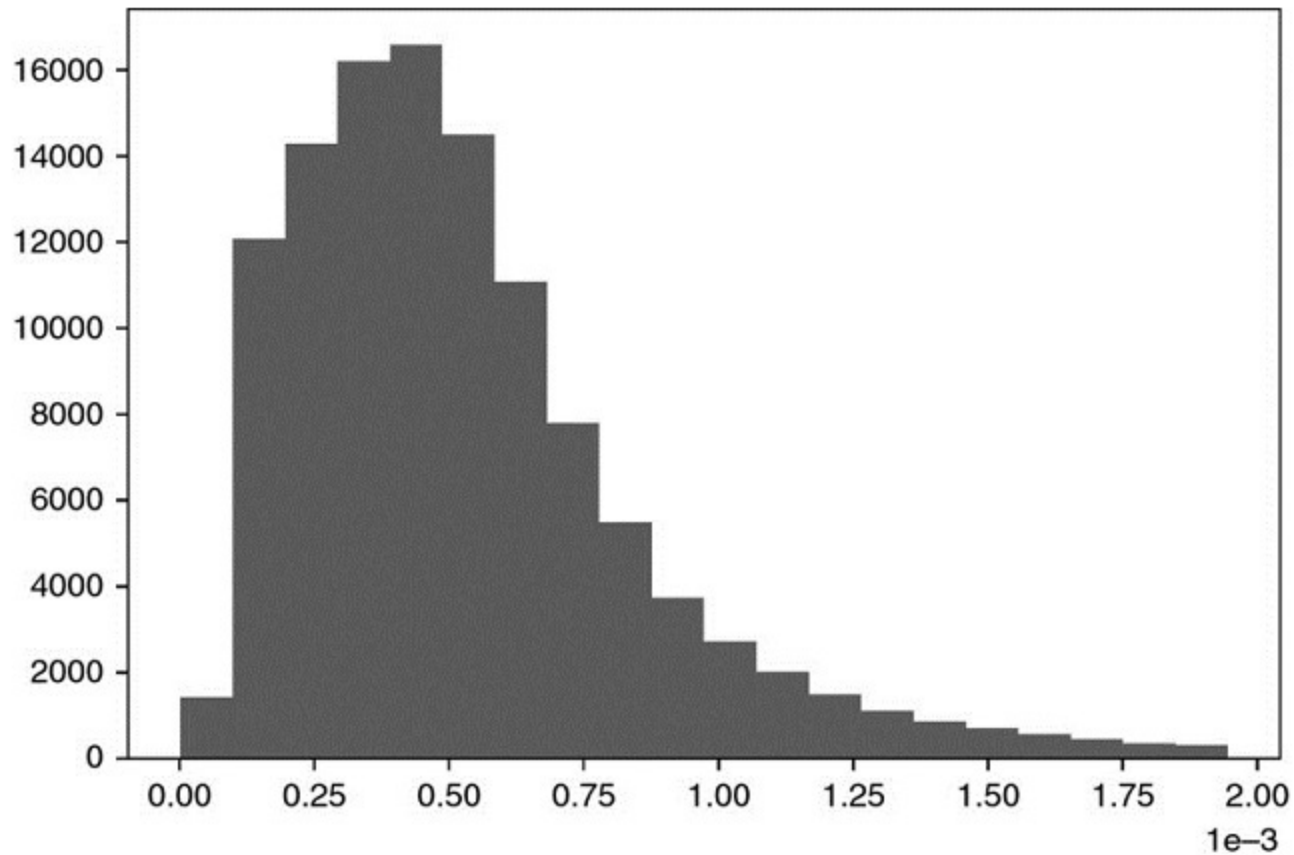


Figure 19.1 Kyle's Lambdas Computed on E-mini S&P 500 Futures

Amihud's λ

- Amihud [2002] computes the daily price response associated with one dollar of trading volume, and argues its value is a proxy of **price impact**.

- Hasbrouck [2009] found that daily Amihud's λ estimates exhibit a high rank correlation to intraday estimates of *effective spread*.

$$|\Delta log p_t^c| = \lambda \sum_{t \in B_\tau} p_t v_t + \epsilon_t$$

where p_t^c is close price ('close' not just calendar time, including economic time as in tick bar), and $p_t v_t$ is volume weighted price.

Hasbrouck's λ

- Estimated λ can be used as a feature approximating the effective cost of trading (market impact)
- Hasbrouck recommends 5-minute time bars for sampling ticks

$$log(p_{i,\tau}) - log(p_{i-1,\tau}) = \lambda_i \sum_{t \in B_{i,\tau}} \left(b_{i,t} \sqrt{p_{i,t} V_{i,t}} \right) + \epsilon_{i,t}$$

where p^c is close price of bar B,

$b_{i,t} \in \{-1, 1\}$ is dummy for buy-or-seller initiated

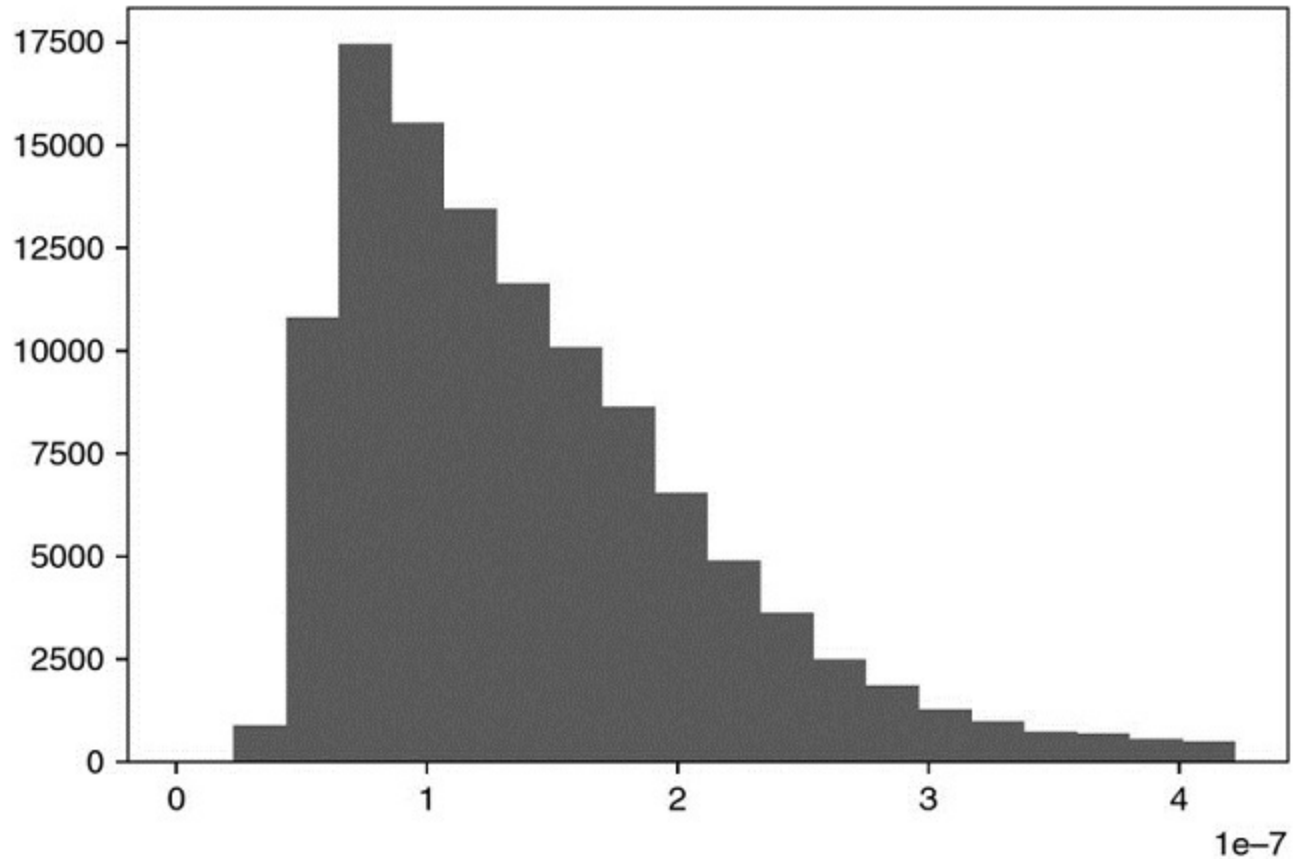


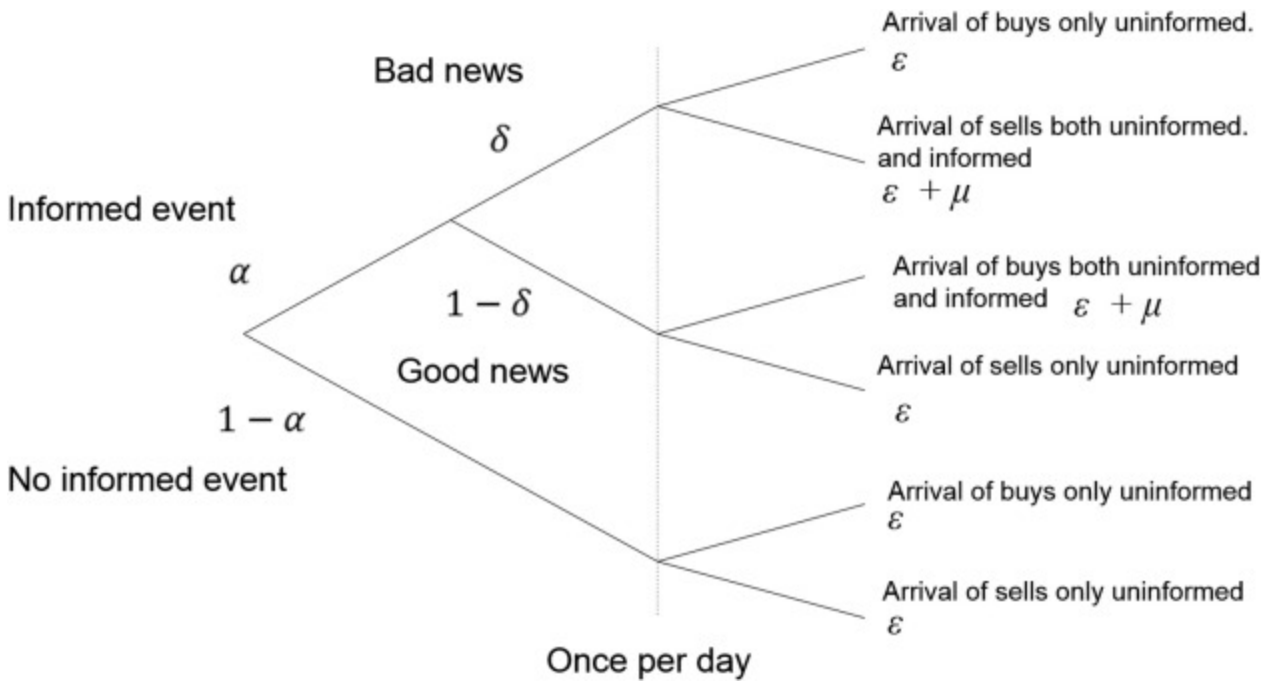
Figure 19.3 Hasbrouck's lambdas estimated on E-mini S&P 500 futures

Third Generation: Sequential Trade Models (STM)

- traders arrive at the market sequentially and independently
- STM is very popular among mm due to identify risk facing by them and to providie guideline how to update dynamically

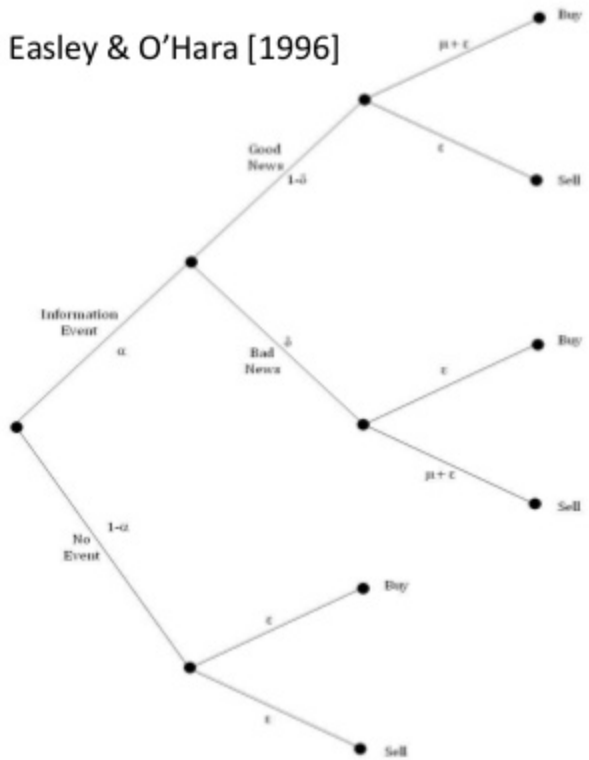
1. PIN (Probability of Information Based Trading)

- Easley et. al.(1996) use trade data to identify the PIN of equity.
- Viewing trading as a game b/w market makers and traders over multiple periods


Bad news
δ
Informed event
α
$1-\delta$
Good news
$1-\alpha$
No informed event
Arrival of buys only uninformed.
ε
Arrival of sells both uninformed.
and informed
$\varepsilon + \mu$
Arrival of buys both uninformed
and informed $\varepsilon + \mu$
Arrival of sells only uninformed
ε
Arrival of buys only uninformed
ε
Arrival of sells only uninformed
ε
Once per day

The PIN Theory

Easley & O'Hara [1996]



PIN estimates the probability that market makers are being adversely selected (i.e., provide liquidity to an informed trader).

$$E[S_i \mid t] = P_n(t)S_i^* + P_b(t)\underline{S}_i + P_g(t)\overline{S}_i$$

$$B(t) = E[S_i \mid t] - \frac{\mu P_b(t)}{\varepsilon + \mu P_b(t)}\left[E[S_i \mid t] - \underline{S}_i\right]$$

$$A(t) = E[S_i \mid t] + \frac{\mu P_g(t)}{\varepsilon + \mu P_g(t)}\left[\overline{S}_i - E[S_i \mid t]\right]$$

$$\Sigma(t) = \frac{\mu P_g(t)}{\varepsilon + \mu P_g(t)}\left[\overline{S}_i - E[S_i \mid t]\right] + \frac{\mu P_b(t)}{\varepsilon + \mu P_b(t)}\left[E[S_i \mid t] - \underline{S}_i\right]$$

$$if\ \delta = \frac{1}{2} \Rightarrow \Sigma = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}\left[\overline{S_i} - \underline{S_i}\right]$$

$$PIN = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}$$

- Notation

  S:security price

  S_B: price when bad news happen

  S_G: price when good news happen

  α: probability of news arrival

  δ: probability of good news

  μ: informed traders arrival rate (poisson)

  ϵ: uninformed traders arrival rate (poisson)

  B_t: bid price set by mm

  A_t: ask price set by mm

$$E[S_1] = (1-\alpha)S_0 + \alpha\big(\delta S_B + (1-\delta)S_G\big)$$

- To avoid losses from informed traders, mm set bid and ask price as following:

$$E(B_t) = E(S_t) - \frac{\mu\alpha_t\delta_t}{\epsilon + \mu\alpha_t\delta_t}\big(E(S_t) - S_B\big)$$
$$E(A_t) = E(S_t) + \frac{\mu\alpha_t(1-\delta_t)}{\epsilon + \mu\alpha_t(1-\delta_t)}\big(S_G - E(S_t)\big)$$

- When $\delta_t = \frac{1}{2}$,

$$E(A_t - B_t) = \frac{\alpha_t\mu}{\alpha_t\mu+\epsilon}\big(S_G - S_B\big) \equiv PIN_t \cdot \big(S_G - S_B\big)$$

  - PIN is, roughly sayin, the ratio of some arrival rate

- How to estimate $\alpha, \delta, \mu, \epsilon$?
    - Using MLE
- Easley, Engle, O'Hara, and Wu(2008) show that

$$E\left(V^B - V^S\right) = \alpha\mu(1 - 2\delta) \approx \alpha\mu$$

where V^B is buy-initiated trading volume.

VPIN (Volume Synchronized Probability of informed trading)

- This model received media attention when it was shown to anticipate the 'flash crash' of May 6, 2010 more than one hour in advance.
- The model adopts volume clock and suggest VPIN as following formulae:

$$VPIN_t = \frac{\sum_{\tau=1}^{n} \left| V_\tau^B - V_\tau^S \right|}{\sum_{\tau=1}^{n} \left(V_\tau^B + V_\tau^S \right)} = \frac{\sum_{\tau=1}^{n} \left| V_\tau^B - V_\tau^S \right|}{nV}$$

where V^B is buy-initiated trading volume.

In sum

- both PIN and VPIN are measures of importance of information based trade
- PIN: one estimate a day , roughly 20 days are enough to infer
- VPIN: based on Volume metric
    - providing a high-frequency metric of the probability that the liquidity provision process may fail.

Additional feature variables from Microstructural datasets

Round-sized trends

- 'mouse' or 'GUI' traders: human traders ordering by clicking buttons, associated with trends (fundamental, belief, conviction)
- 'silicon traders': programmed random trading to disguise footprint

- Rececnt literature found that round-sized(the last digit is zero) trade is more than usual
    - Easley, de Prado, and O'Hara(2016) found that size 10 is 2.9 time more than size 9, 10.9 times(50 vs. 49), 16.8 times(100 vs. 99),32.5 times(250 vs. 249)

- The ratio of $\frac{\text{number of round-sized trades per day}}{\text{number of non-round-sized trades per day}}$ is a candidate of ML features
  - round sized trades associated with trends of human traders
  - lower-than-usual proportion of round-sized trades will be related with short-term view

Cancellation rates, limit orders, market orders

- Eilsler et.al. (2012) argue that different ordering affect on the dynamics of BAS differently
    - market orders, limit ordres and quote cancellations impact on stocks depending on the firm size

- Easley et. al.(2012) argue that large quote cancellation rates may be indicative of low liquidity and very good feature candidates
    - discussed about predatory algorithms (Quote danglers and stuffers, liquidity squeezers, pack hunters)
    - "measuring the rates of quote cancellation,limit orders,and market orders can be the basis for
      useful features, informative of their intentions"

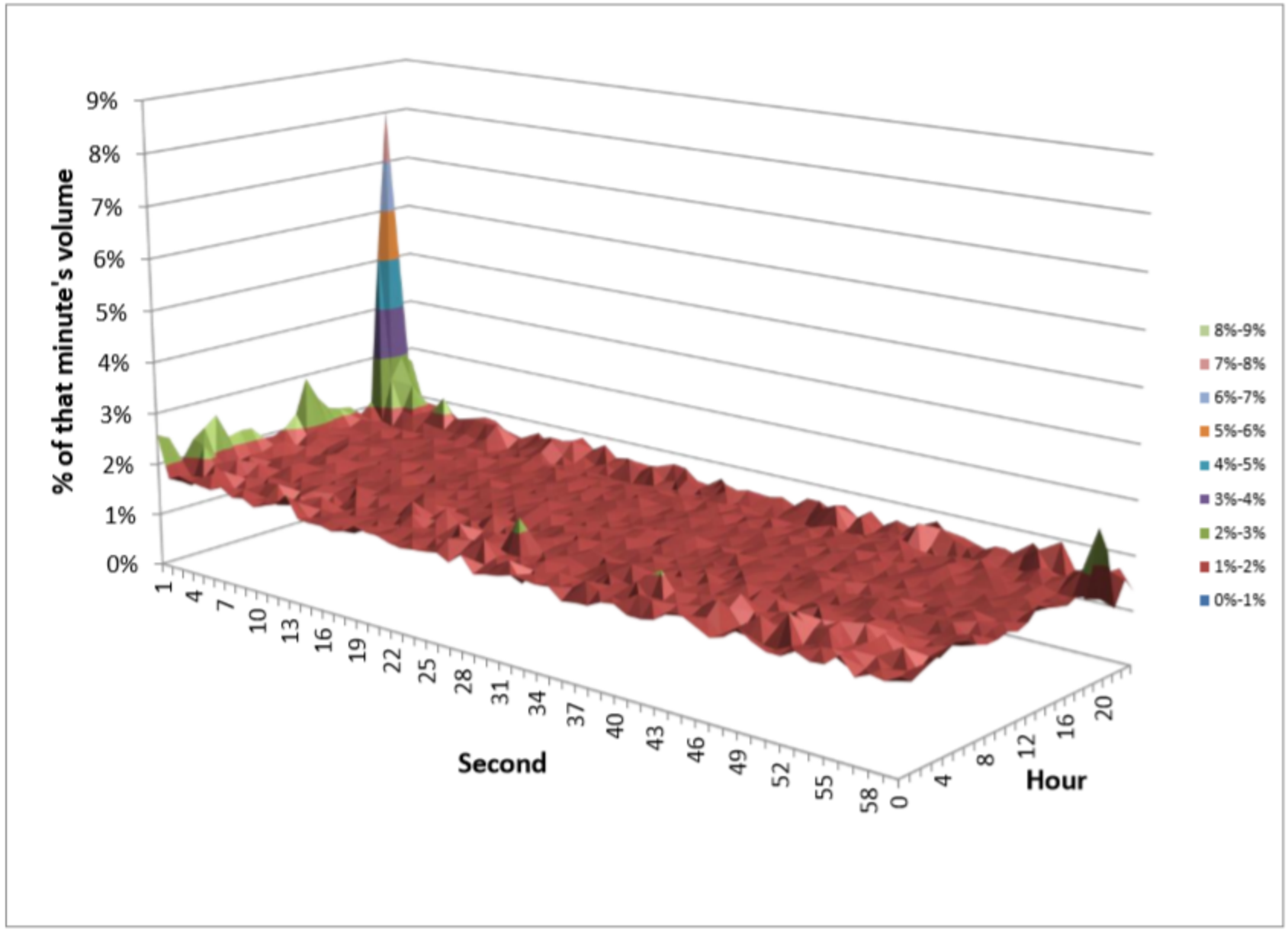
Time Weighed Average Price Execution Algorithms

- Easley et al(2012) show the way how to detect the presence of TWAP
  - Time Weighted Average Price is an algorithm slicing a large order into small ones
  - institutional investors tends to execute large orders in the first second of every minute before the market closes
  - prices are predictable artifacts of the market microstructure
  - HFT(High frequency traders) reacts to information leaked by LFT (low) in order to anticipate their actions

- After seeing the distribution of volume within each minute as the day passes, and search for LFTs executing
  their massive trades on a chronological time-space, it has been found that the largest concentrations of
  volume within a minute tend to occur during the first few seconds, for almost every hour of the day.
    - This is particularly true at 02:00-03:00 GMT (around the open of European equities),
      13:00-14:00 GMT (around the open of U.S. equities) and 20:00-21:00 GMT (around the close of U.S. equities).
    - This is the result of TWAP algorithms and VWAP algorithms that trade on 1 minute slots.

- A useful ML feature may be to evaluate **the order imbalance** at the beginning of every minute,
  and determine whether there is a persistent component.

Exhibit 4 – Percentage of E-Mini S&P500 futures volume traded at each second of every minute



Features from Option markets

- Order flow imbalance is the evidence of the presence of informed traders, hence its intensity of serial correlation would be a good candidate for a ML feature.
- Option Greeks at various maturity and strikes would be a good candidate for features
- Disagreement b/w stock quotes and option price resolved in favor of stock quote, hence, put-call price implied by stock quote shall be a candidate for feature variables in ML

Serial correlation of Signed order flow: Herding or splitting?

- Toth et. al (2011) argue that positive serial autocorrelation a few day is due to order splitting
    - positive autocorrelation is a sign of herding or order splitting, the latter got hands.
- persistency of order flow imbalance to the presence of informedtraders
    - measure the strength of such persistency through the serial correlation of the signed volumes

What is MS information?

feature matrix

$$\mathbf{X} = \{X_t\}_{t=1}^{T} = \Big[\text{VPIN}, \text{Kyles's } \lambda, \text{cancellation rates}, \cdots\Big]$$

Procedure ML

1. Train $(\mathbf{X}, y)$
2. Predict label: $\hat{y}_\tau = E\left[y_\tau | \mathbf{X}\right]$
3. Calculate cross entropy or log-likelihood loss, L_τ
4. Get KDE(kernel density estimator) of $L's$
5. Estimate microstructural information at time *t* as $\phi_\tau = F(L_\tau) \in [0, 1]$

$\Longrightarrow \phi(\cdot)$ **should** be an important feature variables to minimize the risk of crisis- or flash clash-prone period